

Review

A Global Review of Vegetation's Interaction Effect on Urban Heat Mitigation Across Different Climates

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Abstract

The urbanisation process of cities disrupts the natural energy balance and surface radiation, making cities relatively warm. While vegetation has been widely recognised as a key factor in mitigating urban heat, its effectiveness is shaped by interactions with urban morphology, surface cover types, and the background climate. This paper presents a bibliometric analysis of studies examining the role of vegetation in mitigating urban heat, with a particular focus on its interactions within the urban environment across four major Köppen–Geiger climate groups: tropical, arid, temperate, and cold. A total of 130 publications were reviewed, categorised, and analysed according to geographic distribution, study period, and methodological approaches. This review identifies underexplored areas, synthesises key findings, and summarises the most significant results. Vegetation and water bodies emerged as primary contributors to heat mitigation, along with building configuration, wind speed, and shading. Temperate climates were the most frequently studied. Remote sensing was the predominant methodological approach, followed by fixed in situ observations. Meso-scale studies, examining entire cities and their surroundings, dominated in terms of spatial scale. This review offers methodological recommendations for analysing urban vegetation within the context of urban climate research. As climate change intensifies, it is increasingly important to design and implement adaptation strategies that incorporate but are not limited to vegetation. Such strategies are essential to supporting sustainable and resilient urban development in diverse climatic contexts.

Keywords: urban vegetation; heat mitigation; urban climates; cities



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1. Introduction

Humanity faces a significant environmental crisis due to two centuries of rapid economic growth, which has resulted in interconnected global shocks, such as human-induced global warming from greenhouse gas emissions [1]. In response, the 2015 Paris Agreement marked a pivotal step in global climate governance, aiming to limit global temperature rise to below 2 °C above pre-industrial levels, with efforts to restrict this rise to 1.5 °C [2]. These

targets are based on global average temperature increases, a key metric in tracking climate change. Simultaneously, climate change projections show an increase in the frequency of extreme weather events, with urban areas being particularly vulnerable [3]. For this reason, the development of sustainable and healthy cities and transformative changes are required in the coming decades [4], where climatic hazards must be taken into account at a local scale, particularly heat waves, floods, droughts, and storms [5].

Global socioeconomic trends are increasing future exposure to these climatic hazards [6]. Due to population growth, economic development, technological advancements, and energy use, urbanisation is the most dominant anthropogenic phenomenon [7]. The loss of natural habitats, biodiversity, vegetation, and permeable soils in the urbanisation process [8,9] contributes to higher temperatures and pollutant release [10]. Therefore, there is a potential increase in the amount of thermal energy released to the urban climate, where the geographical, climatic, and urban characteristics impact these thermal dynamics [11].

The cause-and-effect relationships within the city–atmosphere system and the multitude of spatial and temporal scales involved in the study of urban’s climates and physical structure are complex [12]. Surface cover typologies, three-dimensional geometry, heat absorption, construction materials, and surface albedo contribute to the formation of diverse microclimates in urban areas [13]. Standard air temperature is a prevalent metric for characterising urban climates and is traditionally contrasted with rural environments. This temperature difference is a well-documented phenomenon known as the urban heat island (UHI), which arises due to the accumulation and re-radiation of heat by buildings, roads, and impermeable surfaces [13]. Recent observational studies on the daily cycle of the UHI have revealed temperature differences influenced by the built environment, nearby rural settings, and prevailing weather conditions [14–16]. Some findings even report lower urban air temperatures compared with rural surroundings, a phenomenon known as the urban cool island (UCI) or negative UHI [17,18].

As early as the 1880s, Hann (1885) [19] observed lower urban air temperatures compared with the countryside in Kolkata, India, noting that both season and geography influence urban–rural temperature differences. UCIs are often observed in arid and semi-arid regions when vegetated or irrigated areas create an oasis effect in urban environments [20]. This cooling effect is studied more during summer than other seasons, as UHIs have relatively little impact on human health in winter, spring, and autumn [21]. Although the UHI effect has been extensively studied, the UCI effect exhibits variability across different urban environments [22,23] and climatic context [24,25].

While UHIs and UCIs describe variations in air temperature, surface temperature patterns (SUHIs and SUCIs, respectively) offer a complementary perspective, particularly valuable in satellite-based assessments [26]. This approach is based on urban and rural horizontal land surface temperatures (LSTs). The main determinants of surface temperature warming in cities are evapotranspiration and convection efficiency [27], which differ markedly from rural conditions. A comparative study by Bechtel et al. [28], examining 50 cities globally, illustrates the substantial variability of SUHI and SUCI phenomena, with vegetated urban areas often exhibiting lower LSTs than adjacent built environments.

Against this backdrop, vegetation in built-up areas serves as a potential heat mitigator, which exerts an influence on reducing LSTs and air temperatures [11,29,30]. However, its effectiveness is also influenced by local climate, soil conditions, water availability, and socio-cultural factors [31]. Identifying how vegetation interacts with urban characteristics to reduce heat is challenging due to significant temperature variability between urban and rural settings. This complexity arises from diverse climatic contexts and the wide range of methodological approaches, study scales, temporal analyses, and data collection methods.

This paper conducts a bibliometric analysis focused on studies exploring vegetation as a heat mitigation strategy in urban environments and its interaction with the built setting. To manage the diversity of studies, classification categories were developed to organise metadata according to geographic patterns, investigation periods, and methodological aspects. To ensure comparability, it is crucial for urban heat mitigation studies focused on vegetation to be well-contextualised within each climatic setting. Therefore, studies were grouped into four primary Köppen–Geiger climate zones: tropical, arid, temperate, and cold.

2. Vegetation in Urban Environments: A General Overview

Vegetation is a fundamental component of urban landscapes, particularly in densely populated areas [32], and many countries have incorporated its monitoring into their political agendas [33]. Vegetation provides various ecosystem services to the urban environment, including surface runoff reduction, flood relief, sustainable drainage, and general aesthetic and well-being enhancements [11]. Notably, it functions as a natural thermostat, helping regulate urban temperatures [34] and improving the quality of life for city dwellers [35].

Urban vegetation provide shading and facilitate evapotranspiration (ET), a process influenced by the availability of water vapour in the air and wind flow [10]. Shading can maintain the air cooler by acting as a solar radiation interceptor, limiting shortwave absorption by urban surfaces and re-radiating heat to the canopy layer [13,36]. However, the alteration in the wind flow due to the surface roughness of the vegetation and buildings canopies alters convective heat exchange [11]. For instance, Arghavani et al. [37] found that low-speed local winds and high levels of air pollutants could offset the benefits of vegetation-induced cooling. In contrast, Yu et al. [38] highlighted that an increase in wind speed would decrease leaf temperature and help transfer the cooling effect to the surrounding areas in a temperate monsoon climate.

Although research generally indicates that the temperature of green spaces is the lowest within an urban area [39], the adjacent built environment can influence their thermal behaviour, potentially leading to a reverse effect due to energy exchange [40]. Analysing the complex interactions between the layout of buildings and vegetation would improve the understanding of the mechanisms that impact the thermal variations exhibited by urban environments [41]. Furthermore, it is evident that the cooling effect of urban greenery does not result from the action of this single factor but rather from complex interaction of numerous elements of the urban environment. These elements potentially include the materiality of the built environment [17], the shading of different elements [42], types of soils [43], wind flows [44], and the distribution of water bodies [45].

3. Methodology

It is crucial to ensure a structured formulation of the research question, the application of methodological filters to retrieve a relevant subset of research, and the specification of a reproducible search procedure [46]. Therefore, a multi-step process was undertaken to select a representative study sample for review and evaluation, consisting of identification, screening, eligibility, and inclusion (Figure 1). This four-step approach follows the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) methodology proposed by [47] for structuring literature retrieval and reporting in systematic reviews.

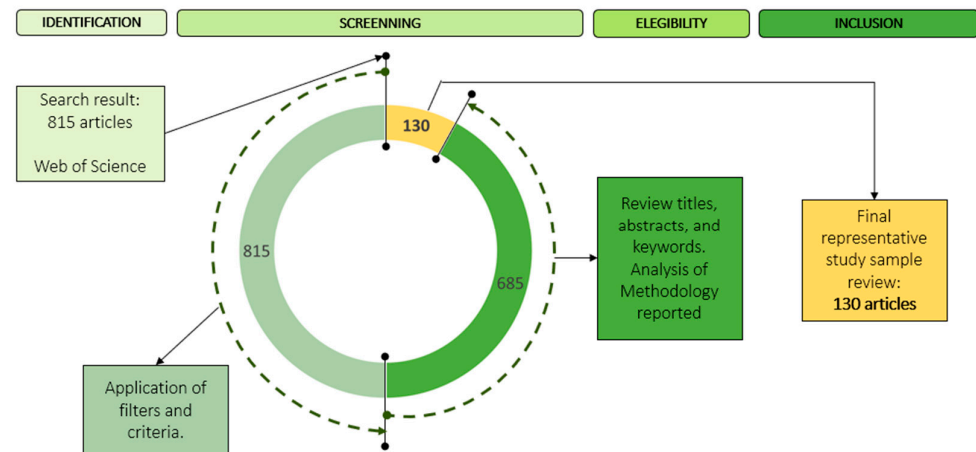


Figure 1. Flow chart methodology of the impact of vegetation on urban heat mitigation in different climate contexts (modified from [47]).

3.1. Identification

An exploratory analysis was carried out as a general search based on selection criteria of scientific publications related to the concepts of this study's objective. A search equation was formulated using keywords and synonyms established in the field of knowledge to identify the relevant articles (Table 1). Moreover, it was necessary to apply exclusion words to retrieve relevant articles. Five thematic groups were identified: urban cool island, urban climate, urban configuration, vegetation, and temperature. Keywords within and between groups were connected by the Boolean operators "OR" and "AND," respectively. The term "local climate zone (LCZ)", a classification system for urban climate proposed by [48], was included in the urban configuration group due to its frequent use in climate-related studies [49]. Likewise, the temperature analysis group aimed to delineate the shape of the UHI/UCI contour, focusing on the urban thermal landscape's details, as these are closely linked to urban construction and human activity [13].

Table 1. Thematic groups of keywords for research search.

Thematic Group	Urban Cool Island	Urban Climate	Urban Configuration	Vegetation	Temperature
Keywords used	"urban cool island *" OR "UCI *" OR "UHI negative" AND "effect" OR "intensity"	"urban" OR "climate" OR "cold" OR "city" OR "urbanization" OR "urban climate"	"morphology *" OR "form *" OR "local climate zone" OR "LCZ *" OR "pattern *"	"green *" OR "park *" OR "forest *" OR "vegetation"	"surface temperature" OR "canopy" OR "land" OR "air temperature" OR "LST"
Excluded words ¹	"transport *" "mobility *" "species *" "agro *" "bio *"				

¹ Excluded words represent a useful way to delimit the field of objective knowledge. * The asterisk detects similar words in singular and plural forms.

The search equation, formulated using the keywords and exclusion words in Table 1, was applied through Clarivate Analytics' Web of Science (WoS) in December 2023. This database was selected as it is the most comprehensive scientific bibliographic resource and aligns with previous research [50,51]. As one of the oldest scholarly databases, WoS offers superior historical coverage compared with others [52], and its interface provides different kinds of data search and download facilities [53]. It is selective, structured, balanced, and provides enhanced metadata, maintaining a wide range of information purposes [54]. These features make WoS a reliable and effective tool for academic research.

3.2. Screening

Once the initial search was carried out, it was necessary to apply filters to retrieve only essential publications from scientific journals (excluding proceedings and irrelevant areas

such as biology, agronomy, medicine, etc.) and in the English language only. Non-peer-reviewed and non-retrievable articles were removed. No temporal cuts were applied.

3.3. Eligibility

The articles were then assessed through a reading process involving titles, abstracts, and keywords, as well as understanding and evaluating the methodology reported. The review sample must be based on the applied air or surface temperature methodology, excluding those articles that studied the urban boundary layer (UBL). Moreover, studies should be based on the central analysis of urban vegetation or similar (parks, gardens, trees along roads, and green roofs and walls).

3.4. Inclusion

In total, 815 articles were identified from the WoS database. However, not all the studies were eligible for inclusion in this study. After the screening and eligibility processes, only 130 articles fulfilled the scope and methodological evaluation criteria and were thus included in this study (Figure 1). It was imperative to account for the variations in the cooling effects of urban vegetation with other urban factors across different seasons, times of day, and climatic zones. This consideration is crucial given that cities with diverse background climates experience distinct thermal environments. Analysing how integrating urban factors with vegetation can enhance and fortify studies related to the cooling effect becomes essential [55,56]. Moreover, it is vital to underscore the regions or cities that have been the focus of these studies, along with the current methodologies employed to compute the UCI or UHI, if applicable.

4. Analysis of Literature Metadata

4.1. Support Software

This study integrated a body of literature by systematically extracting data from a representative sample of studies comprising a manual review and bibliometric analysis. Using Atlas.ti software (version 25.0.1.32924), the manual review facilitated a deeper understanding of specific concepts through coding and analysis [57]. This software program proved beneficial by enabling consistent data extraction from each study using a pre-developed qualitative coding framework. This framework included codes or “tags” for each study’s descriptive characteristics. The software’s advanced features, such as keyword and pattern analysis, relationship charts, and easy access to quotations, enhanced the analysis process and the efficient exploration of relationships and insights [57]. Atlas.ti has been used in studies on renewable energy [58], green infrastructures [59,60], low carbon emissions [61], and sustainability [62,63].

For the bibliometric analysis, VOSviewer software (version 1.16.19) provided a visual map of the studies based on network information, aiding in understanding the connections between the concepts analysed in the manual review [64].

4.2. Manual Review Process

Before analysing each article in the final representative sample, specific categories were established to explore relationships and uncover insights efficiently (Table 2). The categories included geographic patterns, investigation periods, and methodological aspects. Data were organised using Excel spreadsheets. Furthermore, the spatial scale (horizontal range) in the methodological aspects category was based on the classification of urban morphological units by [13]:

- Micro-scale: Roof, wall, building unit, and street or urban canyon.
- Local scale: City block, city centre, residential or industrial zone, and neighbourhood.
- Meso-scale: Built-up area (city) or city plus surrounding countryside (urban region).

- Macro-scale: Cities or regions larger than 100×100 km.

Table 2. Variables of each classification category for the final review of the representative study sample.

Categories	Main Variables (Acronyms)	
Geographic patterns	Reference	
	Country	
	Study area	
	Climate (Köppen–Geiger climate classification)	
Investigation periods	Predominant season or climate	Summer (Su)
		Autumn (Au)
		Winter (Wi)
		Spring (Sp)
		Throughout the year (TY)
		Rainy and dry season (RD)
		Dry season (DS)
		Tropical (Tr)
		Subtropical (STr)
		Hot and wet season (HW)
	Year of Study (YoS)	Year
	Time of day (ToD)	Daytime (DT)
		Nighttime (NT)
		Day- and nighttime (DNT)
Methodological aspects	Temporal analysis method (TAM)	Longitudinal (LN)
		Cross-sectional (CS)
		Future scenarios (FSs)
	Horizontal range (HR)	Macro-scale (Ma)
		Meso-scale (Me)
		Local scale (Lo)
		Micro-scale (Mi)
	Vertical range (VR) (temperature)	Canopy (air) temperature (CT)
		Surface temperature (ST)
	Topic	Urban cool island (UCI)
		Urban heat island (UHI)
		Park cool island (PCI)
		Urban heat and cool islands (UHCIs)
		Water cool island (WCI)
		Urban moisture island (UMI)
		Urban dry island (UDI)
	Methodological approaches (MAs)	Human thermal comfort (HTC)
		Evapotranspiration (ET)
		Spatial configuration (SC)
		Climate (CLI)
		Land surface cover (LSC)
		Socioeconomic factors (SEFs)
		Surface energy balance (SEB)
	Data collection and analysis method (DCM)	Fixed on-site observation (FOO)
		Mobile on-site observation (MOO)
		Remote sensing (RS)
		Experimental (EX)
		Statistical modelling and simulation (SMS)

4.3. Methodological Considerations

To assess the methodological quality of the selected studies, four evaluation criteria were developed, drawing upon key references in the field of urban climatology [65,66]. These criteria were applied to guide the interpretation of results, contextualise the methodological robustness of each study, and identify prevailing gaps in the literature. While a methodological evaluation of the studies is not a central part of this research study and is not included in the results, the use of these criteria was considered important given the inherent complexity of urban climate studies. They served as a guiding framework for comparing study findings and for assessing the coherence and transparency of methodological choices across the literature.

- Criterion I: Clear characterisation of the type of temperature measured.

This criterion verifies whether the study explicitly identifies the type of temperature observed: surface temperature, air temperature, or a measure of both. Distinguishing among these metrics is essential to accurately understanding thermal behaviour in urban environments, given the differences in physical meaning and observational scale. Despite growing consensus regarding the limitations of surface temperature for UHI studies [26,67], its continued use in applied research warrants critical reflection.

- Criterion II: Characterisation of the measurement method.

This aspect evaluates whether the study clearly describes the methodological approach used to obtain temperature data. It includes the identification of the data source (e.g., remote sensing, in situ sensors, and numerical modelling) and relevant technical parameters (e.g., spatial resolution, instrument height, and sampling frequency). Methodological transparency is necessary for cross-study comparison and replicability.

- Criterion III: Explicit identification of the spatial scale.

This criterion assesses whether the study specifies the spatial scale at which observations or analyses were conducted, such as micro-scale (e.g., street canyon), local scale (e.g., neighbourhood), or meso- and macro-scales (e.g., city or metropolitan area). The spatial scale not only influences the manifestation of UHI/UCI phenomena but also affects the choice of measurement and modelling strategies [68,69]. Proper scale definition is fundamental to interpreting results and generalising findings.

- Criterion IV: Explicit identification of the temporal scale.

Temporal scale refers to the timeframe of data collection or analysis, encompassing diurnal, seasonal, or annual cycles. This criterion considers whether the study clearly states the temporal window of observation and whether it accounts for variability in thermal dynamics across different times of day or year. Understanding temporal resolution is vital when interpreting temperature patterns and comparing outcomes across studies.

Together, these four criteria provided a framework for critically evaluating each study's methodological rigour and comparability. Although not employed as exclusion criteria, they were systematically applied during the coding process to assess the internal consistency and analytical depth of the selected literature.

Finally, it was essential to consider the distinction between mitigation and adaptation of urban heat. Mitigation emphasises aspects of the city's physical infrastructure (land cover), while adaptation focuses on human activities, land use, and anthropogenic heat flux [70].

4.4. Climate Classification

The climates analysed were divided into four large groups, equatorial (tropical) climates (A), arid climates (B), warm temperate climates (C), and snow (cold) climates (D),

based on the Köppen–Geiger climatic classification [71]. These climates also include subtypes considering each region’s temperature and precipitation (Table 3). Climate classification is functional in identifying the key findings for urban heat mitigation and future research possibilities. It also facilitates comparing the obtained results with previously published studies with the same (sub-)climates but with different approaches.

Table 3. Description of Köppen–Geiger climatic symbols and criteria for each of the four large groups of climates from [71].

1st	2nd	3rd	Description	Criteria
A			Tropical:	$T_{\text{cold}} \geq 18$
	f		- Rainforest.	$P_{\text{dry}} \geq 60$
	m		- Monsoon.	Not (Af) and $P_{\text{dry}} \geq 100\text{--MAP}/25$
	w		- Savanna.	Not (Af) and $P_{\text{dry}} < 100\text{--MAP}/25$
B	W		Arid:	$\text{MAP} < 10 \times P_{\text{threshold}}$
			- Desert.	$\text{MAP} < 5 \times P_{\text{threshold}}$
			- Rainforest.	$\text{MAP} \geq 5 \times P_{\text{threshold}}$
		h	- Hot.	$\text{MAT} \geq 18$
		k	- Cold.	$\text{MAT} < 18$
C	s		Temperate:	$T_{\text{hot}} > 10$ and $0 < T_{\text{cold}} < 18$
			- Dry summer.	$P_{\text{sdry}} < 40$ and $P_{\text{sdry}} < P_{\text{wwet}}/3$
			- Dry winter.	$P_{\text{wdry}} < P_{\text{swet}}/10$
			- Without dry season.	Not (Cs) or (Cw)
	w	a	- Hot summer.	$T_{\text{hot}} \geq 22$
		b	- Warm summer.	Not (a) and $T_{\text{mon10}} \geq 4$
		c	- Cold summer.	Not (a or b) and $1 \leq T_{\text{mon10}} < 4$
D	s		Cold:	$T_{\text{hot}} > 10$ and $T_{\text{cold}} \leq 0$
			- Dry summer.	$P_{\text{sdry}} < 40$ and $P_{\text{sdry}} < P_{\text{wwet}}/3$
			- Dry winter.	$P_{\text{wdry}} < P_{\text{swet}}/10$
			- Without dry season.	Not (Ds) or (Dw)
	w	a	- Hot summer.	$T_{\text{hot}} \geq 22$
		b	- Warm summer.	Not (a) and $T_{\text{mon10}} \geq 4$
		c	- Cold summer.	Not (a, b, or d)
		d	- Very cold winter.	Not (a or b) and $T_{\text{cold}} < -38$

MAP = mean annual precipitation. MAT = mean annual temperature. T_{hot} = temperature of the hottest month. T_{cold} = temperature of the coldest month. T_{mon10} = number of months in which the temperature is above 10. P_{dry} = precipitation of the driest month. P_{sdry} = precipitation of the driest month in summer. P_{wdry} = precipitation of the driest month in winter. P_{swet} = precipitation of the wettest month in summer. P_{wwet} = precipitation of the wettest month in winter. $P_{\text{threshold}}$ = varies according to the following rules: if 70% of MAP occurs in winter, then $P_{\text{threshold}} = 2 \times \text{MAT}$, and if 70% of MAP occurs in summer, then $P_{\text{threshold}} = 2 \times \text{MAT} + 28$; otherwise $P_{\text{threshold}} = 2 \times \text{MAT} + 14$. Summer (winter) is defined as the warmest (coolest) six-month period between ONDJFM and AMJJAS.

5. Publication Analysis

5.1. Publication History

The first study, published in 1999, examined the influence of geographical factors and meteorological variables on air temperature difference between a park and its built-up surroundings in Göteborg, Sweden [72]. Of the five studies published before 2014, four explored air temperature variations across different urban surfaces, considering urban park design [73] or the time of day [74,75]. The fifth study, published in 2014, used a remote sensing technique to analyse spatiotemporal variations in SUHIs in three megacities: Seoul, Tokyo, and Beijing [76]. From 2015 onwards, the number of publications increased significantly, reaching a peak in 2022 with 28 articles (Figure 2). This growth coincides with global urban sustainability initiatives, including the establishment of the United Nations Sustainable Development Goals (SDGs) in 2015 [77], increased attention to extreme heat

events [78], and heightened awareness of UHI impacts [79]. Among the SDGs, Sustainable Cities and Communities (Goal 11) and Climate Action (Goal 13) are particularly relevant across the reviewed studies.

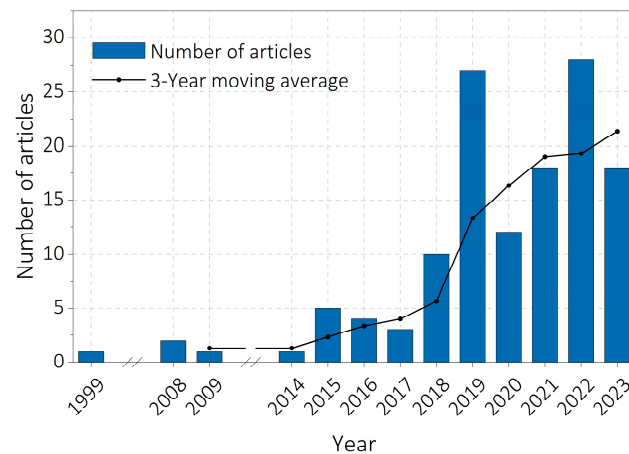


Figure 2. Annual number of articles published according to the methodology applied, with temporal trends shown using a three-year moving average to smooth inter-annual variability and highlight broader patterns.

In the 3-year moving average analysis (black line in Figure 2), short-term fluctuations were smoothed to highlight broader publication trends. The results reveal a clear upward trajectory, particularly after 2014, with a marked acceleration from 2018 onwards. By 2023, the moving average reached 21.13 publications, underscoring the rapid growth of research on urban vegetation, especially regarding heat mitigation and urban climate resilience.

5.2. Geographical Distribution

The studies span 26 countries (see Figures 3 and 4), though research remains geographically uneven. China dominates with 79 studies, followed by India with 13. Conversely, Central and South America and Oceania are underrepresented, with only three studies combined. In 19 countries, only a single study per country meets the inclusion criteria.



Figure 3. The localisation of studies included in this study. Each of the red dots represents one city as case study.

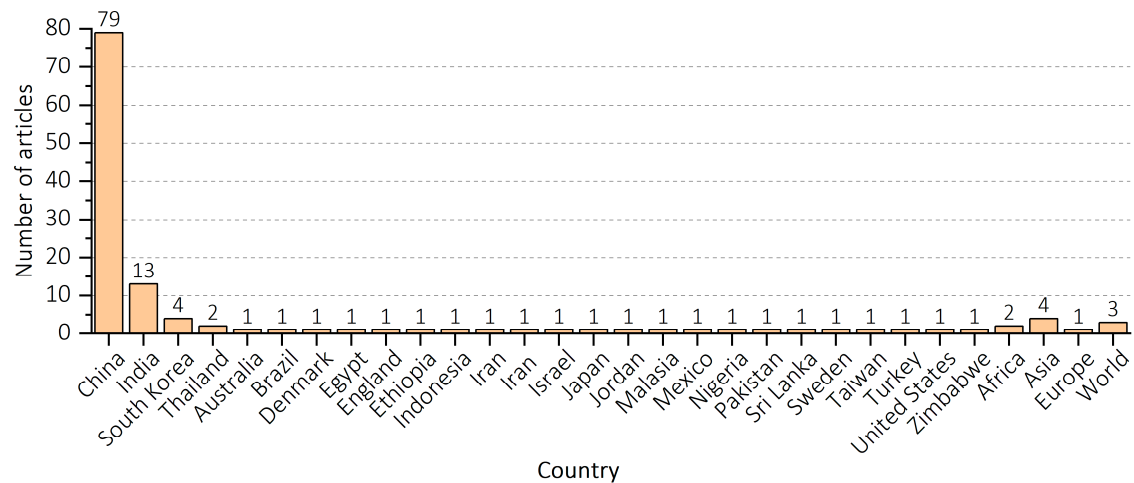


Figure 4. Number of articles covering research on a location in the respective country and continent.

Some studies span multiple cities or regions across continents. As shown in Figure 4, several pan-continental analyses include Africa, Asia, and Europe, along with transnational assessments labelled “World.”

5.3. Climatic Classification

The most frequently studied climate was Cfa (hot summer humid subtropical climate) with 60 studies, primarily in China and 1 in Brazil, followed by Dwa (monsoon-influenced hot-summer humid continental climate) with 23 studies, mainly in China and South Korea. The Aw (tropical savanna climate) classification was examined in 14 studies, mostly in India and Thailand, with some in Ethiopia and Nigeria. Few studies focused on Am (tropical monsoon climate), Csb (warm, dry summer temperate climate), Cwc (cold summer, dry winter temperate climate), Dfa (hot summer humid continental climate), and Dfb (warm summer humid continental climate) regions. Several climate types were not represented in any of the included studies, such as BWk (cold desert arid climate), Csc (cold, dry summer temperate climate), Cfc (cold, humid temperate climate), most D-type climates, and polar climates (E) (Figure 5). This underrepresentation of cold and polar areas may reflect either a lack of urban development or research capacity in these regions [80].

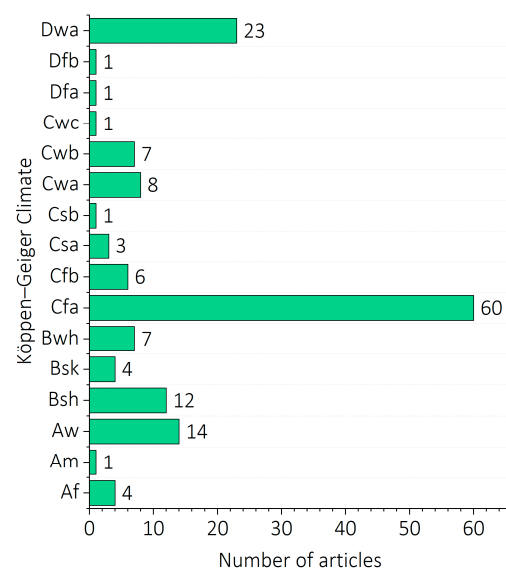


Figure 5. Number of articles covering a specific climate according to the Köppen–Geiger classification (see Table 3).

5.4. Journals Published and Citations

The articles included in this study were published across 49 peer-reviewed journals, covering diverse thematic areas, such as urban and environmental sustainability, climate science and meteorology, remote sensing and geospatial analysis, environmental science and pollution research, and geosciences and earth systems. Figure 6 presents the five most popular journals among the included studies, along with the corresponding number of published articles.

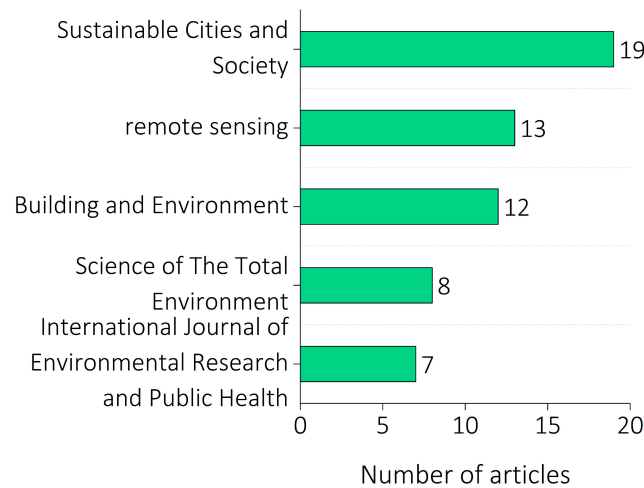


Figure 6. Most popular journals among the included articles in this study.

Based on the WoS database, Table 4 summarises the fifteen most-cited articles in this study. These studies represent significant contributions to the field and are concentrated on high-impact journals such as *Science of the Total Environment*, *Building and Environment*, and *Sustainable Cities and Society*. Highly cited studies frequently use remote sensing, focus on green open spaces, and examine cities in tropical and humid subtropical climates.

Table 4. The top 15 most cited articles according to the WoS web page.

Rank	Times Cited	Title	Authors	Journal	Climate *	DCMs	UFs
1	790	Effects of landscape composition and pattern on land surface temperature: An urban heat island study in the megacities of Southeast Asia	[81]	<i>Science of the Total Environment</i>	Aw and Af	RS	GOSs
2	228	How can urban green spaces be planned for climate adaptation in subtropical cities?	[82]	<i>Ecological Indicators</i>	Cfa	RS	GOSs and WBs
3	217	How can urban blue-green space be planned for climate adaption in high-latitude cities? A seasonal perspective	[83]	<i>Sustainable Cities and Society</i>	Cfb	RS	GOSs and WBs
4	187	A fieldwork study on the diurnal changes of urban microclimate in four types of ground cover and urban heat island of Nanjing, China	[74]	<i>Building and Environment</i>	Cfa	FOO	GOSs, WBs, and SH
5	175	Impacts of urban configuration on urban heat island: An empirical study in China mega-cities	[84]	<i>Science of the Total Environment</i>	Eight climate zones (36 megacities)	RS	GOSs and BC
6	162	Strong contributions of local background climate to the cooling effect of urban green vegetation	[38]	<i>SCIENTIFIC REPORT</i>	Temperate monsoon and Mediterranean climates	FOO and RS	GOSs, WS, HU, and SH
7	132	Spatial regression models of park and land-use impacts on the urban heat island in central Beijing	[85]	<i>Science of the Total Environment</i>	Dwa	RS	GOSs and WBs
8	131	Urban heat island effects of various urban morphologies under regional climate conditions	[86]	<i>Science of the Total Environment</i>	Dwa, Cfa, and Cwb	FOO	GOSs, WS, and BC
9	125	Inter-/intra-zonal seasonal variability of the surface urban heat island based on local climate zones in three central European cities	[87]	<i>Building and Environment</i>	Cfb	RS	GOSs and BC

Table 4. Cont.

Rank	Times Cited	Title	Authors	Journal	Climate *	DCMs	UFs
10	122	Comparison of cooling effect between green space and water body	[88]	<i>Sustainable Cities and Society</i>	Cfa	RS	GOSs and WBs
11	122	Influence of Park Size and Its Surrounding Urban Landscape Patterns on the Park Cooling Effect	[89]	<i>Journal of Urban Planning and Development</i>	Cfa	FOO	GOSs and WBs
12	120	Spatial Analysis of Surface Urban Heat Islands in Four Rapidly Growing African Cities	[90]	<i>remote sensing</i>	Aw, Cfb, and Cwa	RS	GOSs and AL
13	108	Assessing the impact of land use land cover changes on land surface temperature over Pune city, India	[91]	<i>Quaternary International</i>	BSh	RS	GOSs and WBs
14	106	The cooling efficiency of variable greenery coverage ratios in different urban densities: A study in a subtropical climate	[92]	<i>Building and Environment</i>	Cfa	FOO	GOSs and WS
15	103	Spatial structure of surface urban heat island and its relationship with vegetation and built-up areas in Melbourne, Australia	[93]	<i>Science of the Total Environment</i>	Cfb	FOO and RS	GOSs and WS

Abbreviations: data collection and analysis methods (DCMs); urban factors of heat mitigation (UFs); remote sensing (RS); fixed on-site observation (FOO); green open spaces (GOSs); water bodies (WBs); shading (SH); building configuration (BC); humidity (HU); wind speed (WS); albedo (AL); * climates based on Table 3 classification.

5.5. Tendency of Most Recent Works (2023)

Network information processing in VOSviewer for the most recent studies (2023) (Figure 7) shows that the concept of urban vegetation (or green space) has the most robust links with urban heat island, land surface temperature, and city. These connections reflect key terms used in the knowledge field to identify relevant articles, as defined in Table 1. The green space–urban heat island link is well-documented due to its heat mitigation effects [29]. The green space–land surface temperature link represents the most widely applied methodology in recent studies. The green space–city link is a broad connection across studies, with key themes focusing on environmental planning [94], urban ecosystems [95,96], urbanisation [97], climate adaption and mitigation [98], and nature-based solution [99].

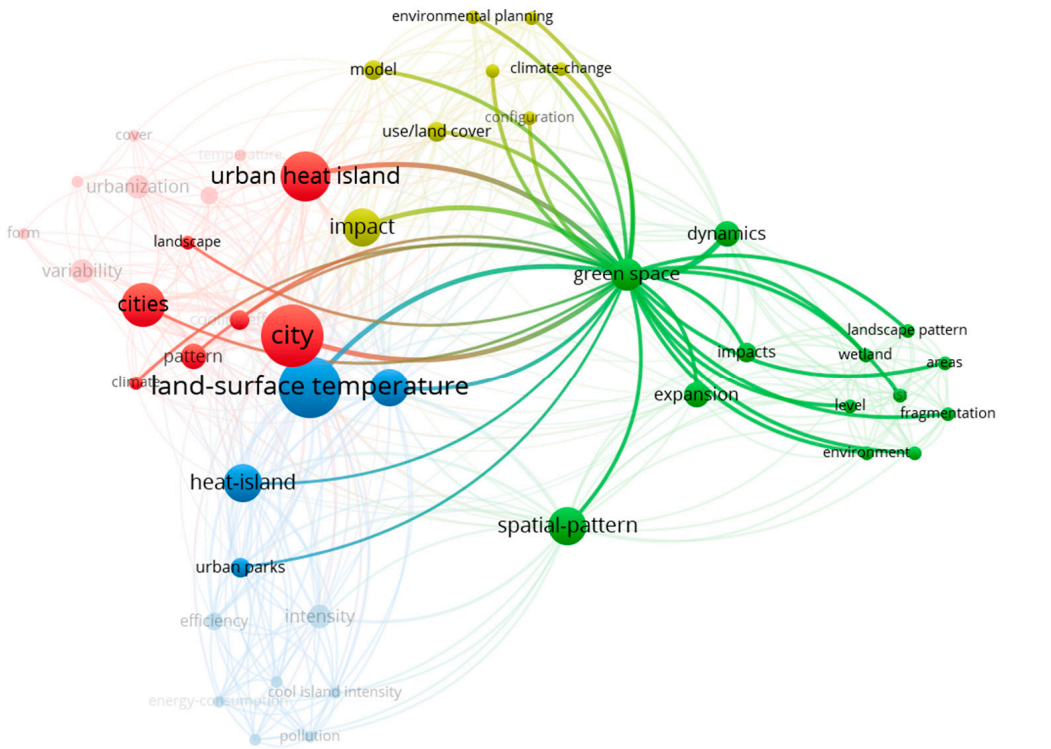


Figure 7. Map of network information for the studies published in 2023.

5.6. Research Areas

The distribution of research areas across the 130 articles retrieved from the WoS database is shown in Figure 8. Most studies were concentrated in environmental sciences, which represented the largest share. This was followed by construction building technology and science technology, both of which accounted for substantial proportions of the literature. Other fields, including biodiversity conservation, computer science, and forestry, appeared less frequently but were recurrent. Overall, the dataset reveals a strong disciplinary orientation towards environmental and climate-related fields, with notable contributions to knowledge fields such as the built environment and remote sensing.

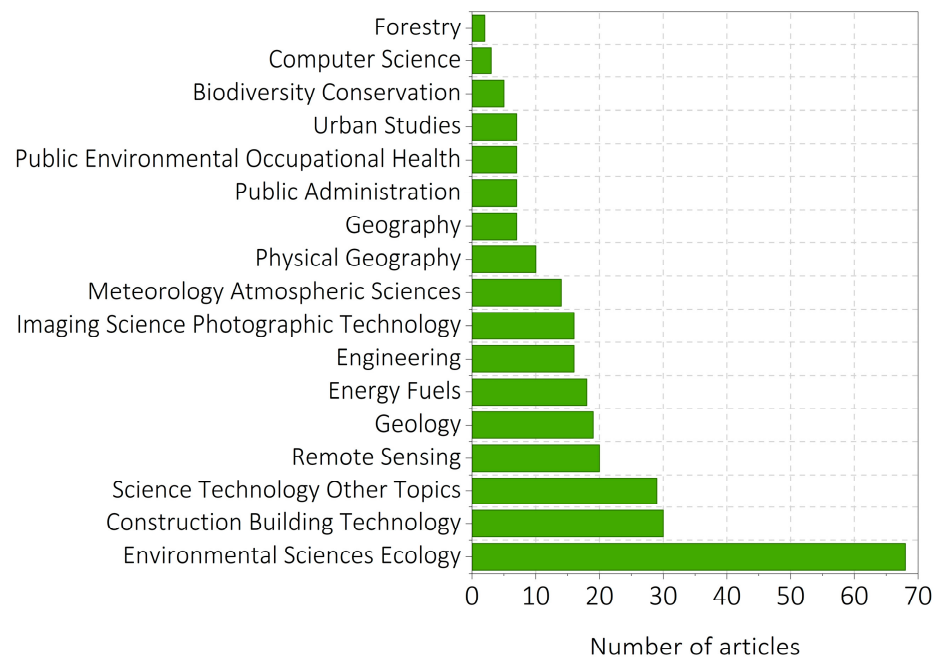


Figure 8. Frequency of research areas represented in the articles, based on the WoS database.

6. Methodological Patterns Across Studies

The spatial scale distribution reveals a predominant focus on the city scale (meso-scale), reflecting broader urban analyses. Regional or global scales (macro-scales) follow, aiming to study several multiple cities or larger geographic regions. In contrast, local and micro-scales were rarely employed, focusing on fine-resolution studies and often tied to urban microclimates (Figure 9A). Regarding the time of day, a significant majority of studies concentrated on daytime and day–nighttime heat mitigation, likely due to its direct impact on human thermal comfort and urban liveability, although these concepts were rarely discussed explicitly in the studies. Additionally, some studies failed to report the time of day, highlighting a methodological limitation in the field (Figure 9B). From a temporal-scale perspective, most studies adopted longitudinal approaches to capturing seasonal or long-term impacts of vegetation. Short-term studies were typically based on snapshot observations or brief monitoring periods (Figure 9C). Data collection methods were dominated by remote sensing, underscoring the central role of satellite-derived thermal data in UHI research. In situ measurements, while valuable for ground truth validation, were less frequently employed due to logistical constraints. Hybrid approaches combining both methods remained limited, suggesting a key opportunity to strengthen methodological rigour (Figure 9D).

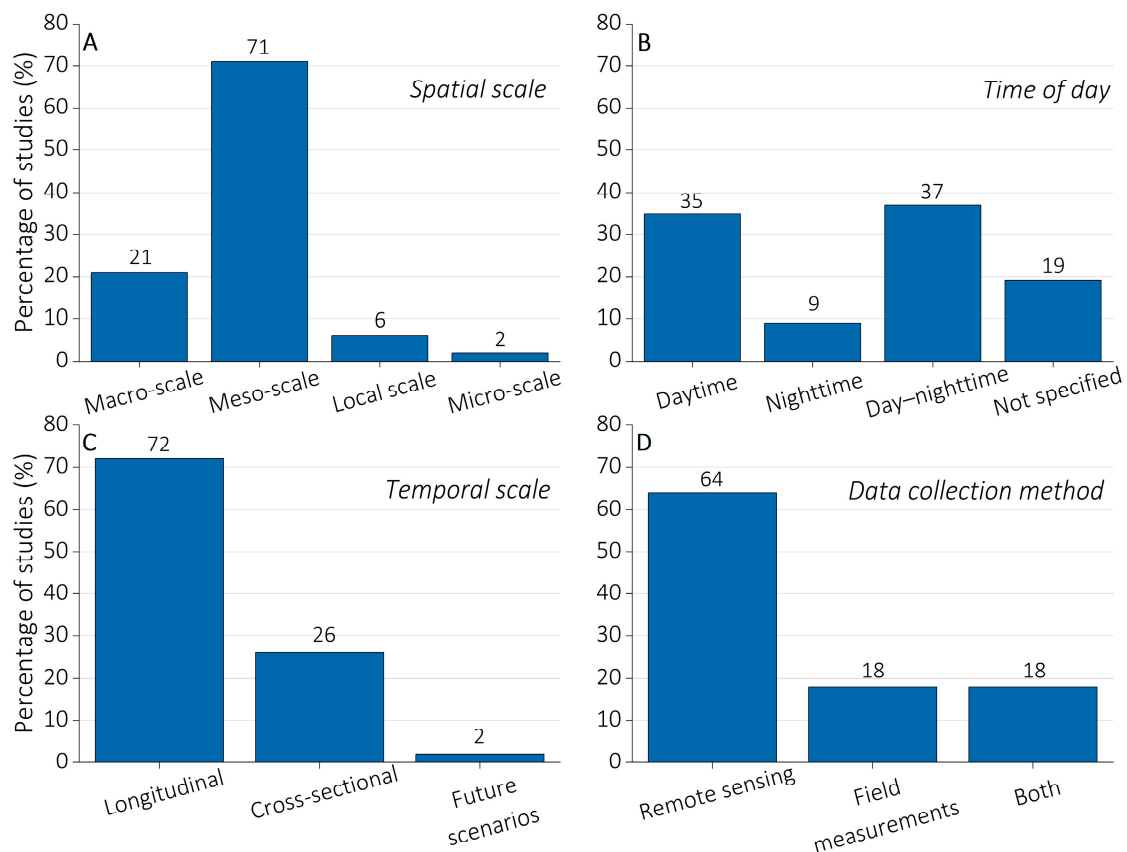


Figure 9. Distribution of studies by (A) spatial scale of analysis, (B) time-of-day approach, (C) temporal scale of analysis, and (D) data collection method.

Together, these patterns underscore current methodological preferences and reveal potential blind spots, particularly the underrepresentation of nighttime assessments, future-scenario modelling, and micro-scale analyses in evaluating the thermal mitigation potential of urban vegetation.

Empirical or statistical models and simulations were the techniques most frequently used in the review studies, such as the random forest model [94] and the boosted regression tree model [100]. Numerical modelling was used to address the limitations of physical and empirical models owing to its scale and scope [101]. Using simulation programs, studies evaluated the impact of different green area shapes and characteristics on the basis of meteorological variables using atmospheric transfer modelling [45] and Reynolds-averaged Navier–Stokes equations [92].

The recent surge in urban heat mitigation simulations focusing on vegetation is likely linked to the development of key modelling tools around the turn of the millennium. ENVI-met was the most commonly used model in the sample. A clear distinction emerged between model scales and the types of analysis conducted. Meso-scale models typically assessed the cooling effects of vegetation in terms of tree cover ratio and green–blue spatial configuration [92,100]. In contrast, local-scale models evaluated the cooling reach of green corridors, particularly those connected to riverbanks or forested areas [41,102]. Although used less frequently, micro-scale models showed that green building façades had a notable cooling effect, extending from the ground level up to 10–20 metres above rooftops and across entire urban blocks [103].

7. Climate-Wise Analysis of Green Space Impacts

This section presents the key findings of studies categorised by climate type, based on the Köppen–Geiger classification. The climate groups considered include A (tropical), B (arid), C (warm temperate), D (cold), and mixed. The most comprehensive results from the reviewed articles are tabulated in Table S1.

Before detailing these results, it is essential to note that potential drivers strongly influence the magnitude and variability of UHI/UCI effects and thus the measured cooling impact of urban vegetation. Understanding the spatiotemporal dynamics of urban systems and their drivers is, therefore, crucial to designing effective climate mitigation and adaptation strategies [68,69]. The analysis of potential drivers, focusing on the spatial distribution, morphology, and composition of urban landscape elements, was particularly valuable in identifying variations at different spatiotemporal scales and in interpreting associations among landscape components [104]. The most prominent drivers identified across studies include the following:

- Biophysical characteristics: Landscape metrics [94,105–109], spatial structures [104,110,111], percentage of different cover areas [45,112,113], and vegetations indices [69].
- Natural factors: Rainfall [39,94], ET [112], radiation [39,69], elevation [113], and aerosol optical depth [69].
- Socioeconomic factors: Gross domestic product [114], population density [69,115], nighttime light [69], socioeconomic activities [113,116,117], and anthropogenic heat flux [118].

7.1. Climate A (Tropical)

In studies conducted in tropical cities such as Lagos, Bangkok, Jakarta, and Denpasar (Table 5; complete metadata in Table S2), dense vegetation and water together yield the strongest cooling effect. Studies report that parks with water bodies produce much greater cooling than green patches alone. For instance, Dutta et al. [119] found an average LST cooling rate of 0.94 °C per 50 m away from a park boundary, with cooling extending up to 200 m, but only in parks that contained enclosed water. Similarly, studies in tropical cities often use the normalised difference vegetation index (NDVI) as a proxy for cooling; Seun et al. [120] showed a tight link between LST and the NDVI, suggesting that even simple regression on the NDVI can predict cooling effects. Moreover, green fragments and corridors still contribute. In Chiang Mai (Aw), over half of urban green space lies in outer zones, meaning that peripheral parks have an outsized influence on mitigating LST [121]. Therefore, key mitigation factors for this climate include the proportion of vegetation and proximity to water bodies [122], as well as physical configurations like patch shape and compactness [98].

While most studies used remote sensing (LST), two combined it with in situ measurements to measure air temperature [123] and urban energy balance terms [124]. One study considered climatic modifiers such as humidity, cloud cover, and wind speed [125]. Notably, the impact of wind speed on urban heat mitigation is most pronounced at night, as it drives significant nocturnal cooling [124]. However, this finding was not consistent across the literature. Conversely, using the LCZ methodology, Kotharkar et al. [125] identified key predictors for urban heat reduction, including distance from the central business district, surface albedo, aspect ratio, and vegetation density ratio.

Overall, tropical cities benefit from high evapotranspiration, and vegetation–water synergies extend cooling well beyond green boundaries, although effectiveness declines beyond a couple of hundred meters. Due to methodological diversity, a direct comparison was limited. Further standardisation and attention to variables such as humidity and wind speed are recommended.

Vegetation cover significantly larger in comparison to impervious surface area (green space-to-impervious surface ratio) is needed urgently to counteract the effects of urban heat [121,126], particularly in megacities [81]. This would improve the quality of life, support sustainable development, and raise public awareness of SUHI issues [127].

Table 5. Overview of studies under climate A. All abbreviations are defined in Table 2 and the Climate column in Table 3.

#	Reference	Country	Climate	HR	VR
1	[123]	Indonesia	Am	Me	CLT and SLT
2	[120]	Nigeria	Aw	Ma	SLT
3	[119]	India	Aw	Me	SLT
4	[125]	India	Aw	Me	CLT
5	[121]	Thailand	Aw	Me	SLT
6	[124]	Malasia	Af	Me	CLT and SLT
7	[126]	Sri Lanka	Af	Me	SLT
8	[122]	India	Aw	Me	SLT
9	[127]	Thailand	Aw	Me	SLT
10	[81]	Asia	Aw, Af	Ma	SLT

7.2. Climate B (Arid)

In arid/semi-arid cities such as Tehran, Amman, Cairo, and parts of India (see Table 6; complete metadata in Table S3), vegetation and irrigated areas generated cooling impacts, but these climates face water-limited conditions. In general, UCIs are more frequently reported in this climate, as daytime UCI effects occur in both summer and winter, particularly in dry semi-arid regions using the LST approach [91,128].

Greater green coverage yields more cooling, but the effect is modest and exhibits anisotropic patterns (cooling effect in different directions). For instance, in Delhi (BSh), the percentage of green space (PLAND) is negatively correlated with LST. However, fragmented green space can raise surface temperatures by decreasing overall evapotranspiration [129]. Thus, large contiguous patches are preferable. For instance, the patch area extends the cooling distance, so larger green patches result in farther cooling reach, but with diminishing returns: beyond 180 m, the extra cooling is $<1\text{ }^{\circ}\text{C}$ [130].

On the other hand, vegetation and water bodies positively influenced the microclimate [131] and had the potential to mitigate heat in dense urban areas [129,132]. However, special attention should be given to street environments. Increasing vegetation along streets enhances cooling through evapotranspiration and shading [133], and it is potentially improved with wind flow [134], but impermeable surfaces are also reduced [135].

Green roof installations can outperform ground greenery for night cooling. In dry Tehran (BSk), Arghavani et al. [37] used high-resolution numerical simulations to examine different green scenarios. They found that surface vegetation provided the lowest diurnal average cooling effect compared with green roofs. Still, its nighttime warming impact was more pronounced than green roofs, highlighting the need for a strategic approach to urban greening.

Although limited in number, studies in this climate suggest that vegetation can mitigate urban heat when carefully selected and managed. Cooling effects were generally smaller and more localised due to low humidity and evapotranspiration. More research is needed to determine optimal irrigation regimes and identify vegetation thresholds for cooling efficacy.

Table 6. Overview of studies under climate B. All abbreviations are defined in Table 2 and the Climate column in Table 3.

#	Reference	Country	Climate	HR	VR
1	[128]	Jordan	BSh	Me	SLT
2	[133]	Zimbabwe	Bsh	Me	SLT
3	[135]	Egypt	BWh	Me	SLT
4	[131]	India	BSh	Me	SLT
5	[91]	India	BSh	Me	SLT
6	[37]	Iran	BSk	Me	CLT and SLT
7	[134]	Pakistan	BWh	Ma	CLT and SLT
8	[136]	India	BSh	Me	CLT and SLT
9	[132]	India	BSh	Me	SLT
10	[129]	India	BSh	Me	SLT
11	[130]	China	BSk	Me	SLT

7.3. Climate C (Warm Temperate)

This climate group includes the most diverse range of studies, covering cities like Melbourne, London, Istanbul, and many cities of China. The magnitude and temporal occurrence of heat mitigation varied significantly across studies. Almost all articles observed seasonal variations, though the highest cooling intensities differed depending on the study's seasonal focus (Table 7; complete metadata in Table S4). The articles focused the urban heat mitigation in summer [22,93,94,106,137], autumn [104,116], or summer and winter [24,82], and some articles analysed all seasons [83,138,139]. Some studies also referred to climate subcategories, such as subtropical [92], hot and wet season [40,140], and dry season [97]. Additionally, several articles examined the cooling effects of urban vegetation based on time of day and heatwave events. For instance, Ouyang et al. [92] assessed different greenery scenarios, finding that a 56% tree cover ratio provided a cooling effect of 0.8 °C during the day and 1 °C during the hottest periods.

Covering the broadest range of research topics from micro- to macro-scale applications, the following analysis is presented across four spatial scales:

- **Micro-scale:** The studies found that vegetation morphology (e.g., trees, gardens, green roofs, and green walls) and surface permeability significantly influence microclimates [92,103,112]. Moreover, key findings included reductions in average temperature, pollution dispersion, and air velocity at the pedestrian level. Peng et al. and Hosseinzadeh et al. [103,141] reported that green walls were generally more effective in lowering air temperature, achieving a maximum cooling intensity of 1.0 °C in simulation studies. Meanwhile, Mandelmlch et al. [141] found that neighbourhood gardens and trees had the most substantial heat mitigation effect, reducing air temperature by up to 3.5 °C, with street orientation also playing a crucial role.
- **Local scale:** Seasonal variability strongly influenced vegetation cooling, with effects more prominent during summer. Vegetation–water synergy was significant in cities with a hot summer humid subtropical climate, where evapotranspiration contributed to greater cooling. Patch configuration, NDVI values, and landscape metrics (e.g., contiguity and shape) were important factors. Urban parks (with water bodies) were highlighted as vital components of highly developed urban ecosystems [112], with cooling effects reaching up to 8.4 °C air temperature reduction [39]. The presence of skyscraper shade further amplified this effect [22]. Additionally, green corridors along rivers facilitated cold air movement, expanding the influence of cooling sources through air passages [102].
- **Meso- and macro-scales:** The studies also consistently found that urban vegetation and water bodies were the most influential factors in mitigating urban heat through

both LST and air temperature approaches. Wind speed and shading played key roles in air temperature reduction. While the earliest study identified wind speed as the dominant factor influencing urban vegetation's cooling effect [72], Huang et al. [75] demonstrated that the combination of water bodies, hill-slope winds, and mountain vegetation significantly lowered urban temperatures. Moreover, shading vegetation along streets reduced total incoming solar radiation [74], impacting heat mitigation. Green spaces and water bodies with irregular shapes created more extensive cooling areas, while fragmented areas were preferable for reducing LST [82,88,137,142–146]. However, seasonal variations must be considered, as green spaces with water bodies exhibited strong cooling effects in summer but showed significant seasonal fluctuations in efficiency [24,25].

In summary, warm temperate cities show high potential for vegetation-based urban heat mitigation, though success is conditional on scale, season, and spatial configuration.

Table 7. Overview of studies under climate C. All abbreviations are defined in Table 2 and the Climate column in Table 3.

#	Reference	Country	Climate	HR	VR	#	Reference	Country	Climate	HR	VR
1	[99]	China	Cfa	Lo	CLT	33	[147]	China	Cfa	Me	SLT
2	[100]	China	Cfa	Me	CLT and SLT	34	[137]	Mexico	Cwb	Me	SLT
3	[97]	China	Cwb	Me	CLT	35	[88]	China	Cfa	Me	SLT
4	[148]	China	Cwa	Me	SLT	36	[149]	China	Cfa	Me	SLT
5	[94]	China	Cfa	Me	SLT	37	[150]	Brazil	Cfa	Me	CLT
6	[112]	China	Cfa	Me	SLT	38	[151]	China	Cfa	Ma	SLT
7	[104]	India	Cwa	Me	SLT	39	[138]	China	Cfa	Me	SLT
8	[94]	China	Cfa	Lo	SLT	40	[143]	China	Cfa	Me	SLT
9	[152]	England	Cfb	Mi	CLT	41	[141]	Israel	Csa	Me, Mi	CLT and SLT
10	[105]	China	Cfa	Me	SLT	42	[92]	China	Cfa	Me	CLT
11	[153]	China	Cfa	Me	CLT	43	[43]	China	Cfa	Me	SLT
12	[154]	China	Cfa	Me	SLT	44	[103]	China	Cfa	Lo and Mi	CLT
13	[44]	Korea	Cfa	Me	SLT	45	[83]	Denmark	Cfb	Me	SLT
14	[45]	China	Cfa	Me	CLT and SLT	46	[155]	China	Cfa	Me	SLT
15	[106]	China	Cwa	Me	SLT	47	[139]	China	Cfa	Me	SLT
16	[156]	China	Cfa	Me	SLT	48	[157]	China	Cfa	Ma	CLT and SLT
17	[107]	China	Cfa	Me	SLT	49	[115]	China	Cfa	Me	CLT and SLT
18	[108]	China	Cfa	Me	SLT	50	[87]	Asia	Cfb	Ma	SLT
19	[22]	Turkey	Csa	Lo	CLT and SLT	51	[158]	China	Cfa	Me	SLT
20	[159]	China	Cfa	Me	SLT	52	[93]	Australia	Cfb	Me	SLT
21	[110]	China	Cfa	Me	SLT	53	[140]	China	Cfa	Me	SLT
22	[18]	China	Cfa	Me	CLT	54	[142]	China	Cfa	Me	SLT
23	[160]	China	Cfa	Lo	SLT	55	[161]	China	Cfa	Me	CLT and SLT
24	[162]	Japan	Cfa	Me	CLT and SLT	56	[82]	China	Cfa	Me	SLT
25	[146]	China	Cfa	Me	SLT	57	[163]	China	Cfa	Me	SLT
26	[39]	China	Cfa	Lo	CLT	58	[40]	Taiwan	Cfa	Me	CLT and SLT
27	[116]	China	Cfa	Me	SLT	59	[117]	China	Cfa	Me	SLT
28	[102]	China	Cfa	Lo	CLT	60	[89]	China	Cfa	Me	SLT
29	[145]	China	Cfa	Me	SLT	61	[164]	China	Cfa	Me	SLT
30	[25]	China	Cfa	Me	SLT	62	[75]	China	Cfa	Me	CLT
31	[144]	China	Cfa	Me	SLT	63	[74]	China	Cfa	Me	CLT
32	[24]	China	Cfa	Me	SLT	64	[72]	Sweden	Cfb	Me	CLT

7.4. Climate D (Cold)

Research on cold climates has primarily focused on cities with a Dwa climate, such as Beijing and Seoul (Table 8; complete metadata in Table S5). Most studies used LST to evaluate cooling from urban vegetation, highlighting significant variation across seasons. The size of green spaces and the presence of water bodies were particularly beneficial in increasing PCI (park cool island) and UCI formation during summer [85,165,166]. How-

ever, Zong et al. [167] noted unexpected effects during heatwaves, such as increased air temperature in green areas.

Vegetation patch shape and edge density also influenced effectiveness; elongated vegetation patches create longer interfaces with surrounding areas, enabling cooler air to be exchanged more effectively, and can also lead to greater heat dissipation beyond the green space [165]. This cooling effect, measured by fractional vegetation cover (FVC), for instance, can reduce temperatures by up to 0.91 °C [113] and reaches an efficiency threshold at 4.5 ha [168]. Conversely, some findings indicate that cooling effects on LST vary significantly by green space size, with cooling intensity demonstrating nonlinear enhancement as vegetation area increases [169–173].

Despite the focused sample (Dwa only), results point to the strategic importance of large, irregularly shaped green patches. Further research is needed to generalise findings across other cold subtypes of climates and better understand anomalous heating effects.

Table 8. Overview of studies under climate D. All abbreviations are defined in Table 2 and the Climate column in Table 3.

#	Reference	Country	Climate	HR	VR
1	[169]	China	Dwa	Me	SLT
2	[113]	China	Dwa	Me	SLT
3	[174]	China	Dwa	Me	CLT
4	[172]	China	Dwa	Me	SLT
5	[166]	China	Dwa	Me	SLT
6	[167]	China	Dwa	Me	CLT
7	[168]	China	Dwa	Me	SLT
8	[175]	China	Dwa	Ma	CLT and SLT
9	[165]	China	Dwa	Me	SLT
10	[171]	Korea	Dwa	Me	SLT
11	[170]	China	Dwa	Me	SLT
12	[176]	Korea	Dwa	Me	CLT
13	[177]	China	Dwa	Mi	SLT
14	[85]	China	Dwa	Me	SLT
15	[178]	China	Dwa	Me	CLT
16	[179]	China	Dwa	Me	CLT
17	[173]	China	Dwa	Me	CLT and SLT
18	[73]	Korea	Dwa	Me	CLT

7.5. Studies Covering Multiple Climates

Some studies compared cities across climate zones, enhancing understanding of how green space performance varies contextually (Table 9; complete metadata in Table S6). For instance, Zhou et al. [98] emphasised spatial heterogeneity, showing that the cooling effect of vegetation on LST depended on patch shape and size in all climate types. Others (e.g., [180]) highlighted challenges in using air temperature across climates due to variations in atmospheric conditions. For instance, wind speed plays a more significant role in mitigating urban heat when background temperatures are low, particularly at night [86].

These multiclimatic studies suggest that certain landscape principles (e.g., minimise built-up areas in small, fragmented urban patches and prioritise extensive green spaces) are broadly effective [111], but outcomes are modulated by local climate, vegetation type, and urban form. Greater methodological standardisation (e.g., use of LCZs or consistent LST thresholds) could enhance comparability.

Table 9. Overview of studies under combination of climates. All abbreviations are defined in Table 2 and the Climate column in Table 3.

#	Reference	Country	Climate	HR	VR
1	[69]	China	-	Ma	SUT
2	[181]	China	-	Ma	CLT and SUT
3	[109]	China	Dwa, Cfa, and Aw	Ma	SLT
4	[182]	India	Aw and BSh	Ma	SLT
5	[96]	China	-	Ma	SLT
6	[95]	Ethiopia	Cwb, Aw, BSh, and Cwa	Ma	SLT
7	[109]	China	Dwa and Cfa	Me	SLT
8	[23]	World	-	Ma	CLT and SLT
9	[42]	Africa	BWh, Cwb, Af, and Aw	Ma	SLT
10	[17]	India	Aw, BWh, BSh, Cwa, Cwb, and Cfa	Ma	SLT
11	[183]	India	BSh, BWh, Aw, and CWA	Ma	CLT and SLT
12	[86]	China	Dwa, Cfa, and Cwb	Ma	CLT
13	[41]	China	Dwa and Cfa	Lo	CLT
14	[184]	World	BWh, BSh, ET, Csa, BSk, Cwb, and Csb	Ma	SLT
15	[118]	China	-	Ma	SLT
16	[90]	Africa	Aw, Cfb, and Cwa	Me	SLT
17	[84]	China	Eight climate zones	Ma	CLT and SLT
18	[185]	China	-	Ma	CLT and SLT
19	[180]	United States	Dfb, BWh, Cfa, Dfa, and BSk	Me	CLT
20	[186]	Asia	-	Ma	SLT
21	[187]	Iran	-	Ma	SLT
22	[38]	World	-	Ma	CLT and SLT
23	[111]	Europe	-	Ma	SLT
24	[114]	China	-	Ma	SLT
25	[76]	Asia	Dwa, Cfa	Ma	CLT and SLT

8. Discussion of the Current State of the Role of Vegetation in Urban Heat Mitigation

This discussion is structured into three subsections: The first explores the implications of urban vegetation for heat mitigation across different climatic contexts and proposes future research directions. The second addresses the methodological challenges in assessing urban heat mitigation. The third emphasises the need for context-specific, holistic, and parameter-based evaluations to guide effective mitigation strategies.

8.1. Urban Vegetation, Climatic Contexts, and Future Directions

This study examined the role of urban vegetation in mitigating urban heat across tropical, arid, temperate, and cold climate zones. Notably, certain climatic regions, particularly arid and tropical zones (Köppen climates A and B), were underrepresented in the literature. This limited attention may be due to a lack of research attention and funding to investigate cities in these climates or the inherent challenges posed by aridity, such as water scarcity and low vegetative cover [188,189]. Figures 3 and 4 further substantiate this gap, revealing a lack of studies in regions such as Latin America, Southeast Asia, Africa, and Oceania.

Given the unequal global warming trajectories [190], the underrepresentation of certain regions in urban heat research is concerning. Rapid urbanisation in the Global South, particularly in Latin America, necessitates greater scholarly attention due to its projected population growth and heightened vulnerability to climate impacts [191]. Water scarcity,

intensified by projected changes in rainfall patterns, presents a significant constraint in implementing vegetation-based mitigation strategies in many regions [192–194]. Consequently, greater emphasis on passive cooling strategies that incorporate building morphology, ventilation, and shading in conjunction with vegetation is needed. While studies have assessed urban passive cooling (e.g., [195–197]), few have evaluated synergies between vegetation and other urban parameters.

In colder climates (climate D), vegetation's cooling effect may be detrimental in winter when the UHI effect contributes positively to thermal comfort and energy savings [198]. In contrast, during summer, cooling strategies remain necessary. This seasonal duality underscores the importance of designing mitigation approaches that consider temporal variability (day vs. night, summer vs. winter, etc.).

Although vegetation and water bodies consistently demonstrate cooling potential, the effectiveness of mitigation strategies is strongly modulated by synoptic weather conditions, including wind, humidity, and boundary layer height [23,66]. Vegetation not situated within large parks, such as street trees or small green patches, can offer significant cooling benefits under specific meteorological conditions, especially in arid climates during windy daytime periods [86,174]. However, cooling efficacy is contingent on wind direction [102] and the orientation of urban canyons [103]. Furthermore, low vegetation may be more effective in dry atmospheres than in humid ones, while rooftop greening tends to influence pedestrian-level temperatures in low-rise areas, but less so in high-rise neighbourhoods. Considering the horizontal building configuration, dispersed, irregularly shaped building patterns with vegetation were found to reduce UHI intensity effect [107,108]. Vertically, a mix of high- and low-rise structures, particularly in combination with vegetation, promotes ventilation and cooling [22,42]. Studies using the local climate zone (LCZ) classification found that “open low-rise” configurations exhibited the most effective cooling [133].

Many studies rely on LST obtained by remote sensing techniques, which may not accurately represent near-surface air temperatures that are more relevant for human exposure and thermal comfort. It raises a challenge in urban areas subject to extreme heat that are not evenly distributed spatially. While there is broad agreement in the literature that hotspots occur in densely built-up areas with limited vegetation and/or water, more robust assessments using air temperature data from strategically selected locations could strengthen heat–health protection strategies. However, given the limitations of installing extensive weather station networks, remote sensing, widely applied in the studies reviewed here (Figures 8 and 9D), offers a valuable means of capturing a spatial snapshot of LST over large areas and can serve as a proxy for air temperature, although correlations may weaken under unstable (windy) conditions. Despite the clear distinction between LST and air temperature, targeting reductions in high surface temperatures remains an important goal, as such areas typically coincide with elevated air temperatures and higher solar radiation absorption, directly affecting human thermal comfort [199].

Areas with high pedestrian activity, such as public transport interchanges, recreational spaces, outdoor strips, schools, and pedestrian thoroughfares, should be prioritised for heat mitigation to improve thermal comfort for large segments of the population [29]. Shading offers a promising solution yet remains relatively underexplored in the reviewed studies. While vegetation shading can enhance latent heat fluxes, building shading, particularly from high-rises with reflective roofs, has shown a more pronounced cooling impact on LST [75,135,169]. Further investigation is warranted to evaluate the effects of shading in different seasonal, morphological, and climatic contexts, particularly in arid zones where evapotranspiration may be limited. It is important to note that although extensive evaporation can reduce air temperature, it simultaneously increases humidity and may not necessarily enhance thermal comfort. In many climates, especially in arid ones (climate

B in this study), vegetation-based strategies are ineffective without irrigation, making vegetation, irrigation, and shading strategies closely interdependent.

Urban surface materials also play a crucial role; albedo, thermal inertia, and emissivity influence surface and air temperatures. Remote sensing has shown high-albedo materials to be effective in reducing daytime LST, particularly in arid and warm temperate climates [17,184]. However, interactions between surface materials and vegetation are complex. For example, asphalt combined with trees showed lower air temperatures during the day but led to higher nighttime temperatures due to heat release [75].

8.2. Beyond UHI Magnitude Metrics

Across all climates, vegetation was consistently associated with reduced UHIs and improved environmental quality, including pollutant removal and psychological well-being [200–202]. Nevertheless, a disproportionate focus on UHI/UCI magnitude persists in the literature. This metric, while common, may obscure more nuanced dynamics of heat exposure. Air temperature is influenced by boundary layer dynamics, advection, and radiative processes, factors not fully captured by surface measurements [68,203]. Urban morphology, socio-demographics, and regulatory contexts also affect thermal profiles, complicating comparative analysis across studies.

Crucially, this study avoided using the term “UHI mitigation” without sufficient empirical justification. Instead, it emphasised direct assessments of heat reduction attributable to vegetation; therefore, we highly recommend more nuanced frameworks such as those proposed by Martilli et al. (2020) [204], based on LCZ comparability. To enhance reliability, future research should prioritise comprehensive assessments that integrate vegetation with other urban parameters, particularly when comparing urban and rural environments, as the LCZ framework classification, and considering potential drivers (see Section 7). Where computational resources allow (e.g., through ENVI-met, WRF [Weather Research and Forecasting], BEP, or BEM.ENVI-met), we recommend evaluating urban vegetation both in isolation and in combination with other mitigation strategies using the other urban parameters discussed above. This dual approach enables a clearer understanding of the individual cooling effectiveness of each strategy while also enabling the exploration of potential interactions as demonstrated by Stepani and Emmanuel (2022) [205], who showed that green interventions, using ENVI-met simulations, could substantially cool the Jakarta metropolitan area. Furthermore, the accurate modelling of urban vegetation requires integrated soil–vegetation–atmosphere schemes coupled with urban models that account for building and surface interactions, with a focus on thermal comfort, as shown by Du et al. (2022) [206], who employed a coupled WRF–BEP/BEM model with a high-resolution LCZ map based on the WUDAPT methodology.

8.3. Methodological Limitations and Recommendations

Urban systems are characterised by a high degree of heterogeneity and complexity, making it difficult to isolate the drivers of UHI effects or the effectiveness of specific mitigation strategies [207]. The case-specific nature of many studies precludes generalisation, especially when climatic, geographical, and morphological conditions vary widely. Thus, future work should adopt holistic, climate-sensitive frameworks for evaluating urban heat mitigation. Mitigation strategies should be assessed not in relation to rural baselines, but according to the intrinsic thermal characteristics of urban surfaces, such as materiality, geometry, and human activities [204]. Many studies rely on large datasets with limited spatial representativeness or temporal control. As Stewart (2011) [66] argues, a smaller, controlled dataset may yield more robust insights than larger, less rigorous ones. Similarly,

site representativeness is more important than sample size when seeking to characterise spatial variability in urban heat.

The urban–rural dichotomy also becomes problematic in highly urbanised regions where city boundaries are blurred, as in the “desakota” regions described by [208]. The rural context itself may no longer provide a suitable baseline. Urbanisation, increasingly understood as a planetary process [209], necessitates a broader conceptual framework that encompasses both intra- and extra-urban transformations. To improve methodological rigour, urban climatology should incorporate four core parameters: structure (geometry), cover (surface characteristics), fabric (materials), and metabolism (anthropogenic activities) [13]. Yet many reviewed studies did not isolate these parameters, weakening claims about vegetation’s cooling efficacy. The LCZ system provides a robust alternative to traditional land use/land cover classifications by accounting for morphological characteristics. Its wider adoption is strongly recommended.

Finally, LST and air temperature should be analysed in tandem, as they are governed by different processes but linked through land–atmosphere interactions. Greater attention should also be given to intra-urban variability, anthropogenic heat sources, and meso-scale circulations such as sea breezes and mountain–valley winds, which can modulate local thermal dynamics [204]. Furthermore, it is essential to assess the combined effects of vegetation and other urban parameters (analysed here) as mitigation strategies on air temperature, as this variable plays a critical role in evaluating human thermal comfort.

9. Conclusions

This study synthesised relevant knowledge on the role of urban vegetation in mitigating urban heat across different climatic contexts, with particular attention to spatial, temporal, and methodological patterns. Using the Web of Science database with a reproducible search query and eligibility criteria, 130 publications have been identified for reviewing, categorised, and analysed according to their publication year, country, climate, research topic, methodology, keywords, citations, and publication channels. The findings confirm that vegetation consistently contributes to urban cooling, although its effectiveness varies considerably depending on regional climate, urban morphology, meteorological conditions, and surface characteristics. Incorporating urban parameters, such as structure (e.g., buildings and trees), coverage (e.g., vegetation, water, material, and soil), variable climatic factors (e.g., wind), and a factor that arises because of geometry (e.g., shading), yields more accurate results to mitigate urban heat.

Understudied regions, especially in the Global South and in arid and tropical climates, require greater scholarly attention, given their vulnerability to intensifying heat and rapid urbanisation. Moreover, vegetation-based strategies must be tailored to local constraints such as water availability, seasonal variability, and socio-ecological conditions. Methodologically, the literature remains dominated by LST assessments and generalised urban–rural comparisons. This limits the depth and transferability of findings. A shift towards more nuanced, climate-sensitive frameworks, such as the LCZ method and intra-urban analyses, is essential. Future research should prioritise comprehensive approaches that account for the complex interplay among biophysical, morphological, and anthropogenic factors.

Ultimately, designing effective heat mitigation strategies requires both technical refinement and contextual understanding. Strengthening empirical research in under-represented climates, improving methodological coherence, and integrating vegetation within broader urban systems will be critical to advancing sustainable and equitable urban climate resilience.

To summarise, the specific findings are as follows:

- Most studies examined Cfa climates (60 articles), primarily in China, followed by Dwa (23), mainly in China and South Korea. Aw (14) was the third most studied climate type, with research concentrated in India and Thailand.
- The majority of the studies used remote sensing measurements to obtain LST, employed in 86 of 130 articles. Air temperature was assessed through fixed and mobile on-site measurements in 25 studies, while 21 studies combined both techniques (remote sensing and on-site observations). Only one study used experimental methods to evaluate mitigation strategies (e.g., green roofs, green walls, and trees) for reducing the urban heat island (UHI) effect.
- In terms of spatial scale, meso-scale studies (91 articles) were the most common, analysing built-up areas or entire cities with their surroundings. Macro-scale studies (28) compared cities or regions, while local-scale studies (8) focused on neighbourhoods or city blocks. Micro-scale studies (3) analysed individual urban elements such as roofs, walls, building units, streets, or urban canyons.

The insights gained from this systematic review and meta-analysis provide researchers with a clearer understanding of the current landscape of urban climate research. By enabling comparisons with previous studies, this work reveals valuable connections and learning opportunities. The statistical analysis and synthesis presented highlight both central themes and secondary issues related to vegetation-based cooling strategies, deepening our understanding of urban climates across different regions. This information enables researchers to easily locate relevant studies and identify critical gaps for future investigation.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/urbansci9090361/s1>, Table S1: Summary of the relevant findings of most studies reviewed according to the criteria established in the methodology, Table S2: Overview of studies under climate A. All abbreviations are based on Table 2 and the Climate column in Table 3 of the manuscript, Table S3: Overview of studies under climate B. All abbreviations are based on Table 2 and the Climate column in Table 3 of the manuscript, Table S4: Overview of studies under climate C. All abbreviations are based on Table 2 and the Climate column in Table 3 of the manuscript, Table S5: Overview of studies under climate D. All abbreviations are based on Table 2 and the Climate column in Table 3 of the manuscript, Table S6: Overview of studies under combination of climates. All abbreviations are based on Table 2 and the Climate column in Table 3 of the manuscript.

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Abbreviations

The following abbreviations are used in this manuscript:

Au	Autumn
CLI	Climate
CS	Cross-sectional
CT	Canopy (air) temperature
CUCI	Canopy (air) urban cool island
DNT	Day- and nighttime
DT	Daytime
DS	Dry season
ET	Evapotranspiration
EX	Experimental
FOO	Fixed on-site observation
FS	Future study
HTC	Human thermal comfort
HW	Hot and wet season
LN	Longitudinal
Lo	Local scale
LSC	Land surface cover
Ma	Macro-scale
MAs	Methodological approaches
Me	Meso-scale
Mi	Micro-scale
MOO	Mobile on-site observation
NT	Nighttime
PCI	Park cool island
RD	Rainy and dry season
RS	Remote sensing
SC	Spatial configuration
SEB	Surface energy balance
SEF	Socioeconomic factor
SMS	Statistical modelling and simulation
Sp	Spring
ST	Surface temperature
STr	Subtropical
Su	Summer
SUCI	Surface urban cool island
Tr	Tropical
TY	Throughout the year
UBL	Urban boundary layer
UCI	Urban cool island
UHCI	Urban heat and cool islands
UHI	Urban heat island
UMI	Urban moisture island
WCI	Water cool island
Wi	Winter
WoS	Web of Science

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